Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***
Code: 101603 | ***Subject: Design of Steel Structure*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) If wind or earthquake load is acting, how much more stress can you allow in bolts or rivets as per IS 800:2007?
(i) 15% more
(ii) 33% more.
(iii) 20% more
(iv) 25% more

(b) What is the factor of safety typically used in the Working Stress Method for steel structures?
(i) 1.67
(ii) 1.5
(iii) 2
(iv) 1.76

(c) In Limit State Design, the partial safety factor for materials is usually taken as
(i) Equal to1
(ii) Greater than 1
(iii) Equal to 0
(iv) Less than 1

(d) Which section, according to IS 800:2007, reaches yield stress in the extreme fibre but local buckling prevents it from developing plastic moment capacity?
(i) Plastic
(ii) Slender
(iii) Semi-compact
(iv) compact

(e) In an eccentric bolted connection, the bolt group is subjected to
(i) Only shear
(ii) Only tension
(iii) Combined shear and tension
(iv) Only compression

(f) Which type of bolt suitable to withstand stresses that changes the direction repeatedly.
(i) Black bolt
(ii) Turned and fitted bolt
(iii) Ordinary unfinished bolt
(iv) High strength bolt

(g) Which factor influences the size of a fillet weld most?
(i) Length of the weld
(ii) Thickness of plates being joined
(iii) Distance between welds
(iv) Quality of steel

(h) Shear lag effect in a tension member is significant when
(i) Connection is at the centroid directly
(ii) Member is made of only plates
(iii) A part of cross-section is not connected directly
(iv) Holes are drilled away from centroid

(i) The effective length of a column fixed at one end and free at the other is
(i) L
(ii) 2L
(iii) L/2
(iv) L/√2

(j) Block shear failure in plate girders typically occurs
(i) In the flange
(ii) At web-flange junction
(iii) At the connection between web and stiffener
(iv) Along a plane through the bolt line and edge of the plate

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Q.2 (a) What are the advantages and disadvantages of steel structures? How you classified connection provided in a steel structure? [7]

(b) Sketch and briefly explain any three failure patterns of bolted connections. [7]

Q.3 (a) Design a double cover butt joint between the two plates of width 300 mm, if the thickness of one plate is 18 mm and the other is 10 mm. The joint has to transfer a factored load of 480 kN. The plates are of Fe 410 grade. Use M20 bolts of grade 4.6. Cover plates are 8mm thick. [7]

(b) Write three advantages and disadvantages of welded joints? [7]

Q.4 (a) Explain the various types of loads to be considered and load combinations to be considered in the design of steel structures. [7]

(b) Write the specifications provide by IS 800:2007 for fillet welds and butt welds. [7]

Q.5 (a) Explain the purpose of lug angles in tension member design. [7]

(b) Design a truss diagonal subjected to a factored tensile load of 180 kN. The tension member is connected to a 20mm thick gusset plate with one line of M20 bolts of grade 4.6 [7]

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Q.6 (a) Write the detailed procedure for design of a bolted gusset base for a steel column. [7]

(b) Define lateral-torsional buckling with a neat sketch and also write the assumptions. [7]

Q.7 (a) Design a column to support a factored load of 1 200 kN. The length of the column is 4 m and both the ends are hinged. Use steel of grade Fe 410. [7]

(b) Illustrate the different elements of a welded plate girder with sketches. [7]

Q.8 An ISWB350@558.19 N/m has been used as a simply supported beam over a span of 6.5m. The beam has a cover plate of 200 mm width and 8 mm thickness on top and bottom flanges. Determine the safe uniform load that the beam can carry in flexure if the compression flange is restrained against lateral buckling. [14]

Q.9 Write short note on ***any four*** of the following: [3.5 x 4 = 14]

(a) Limit state design method Versus working stress design method

(b) Factors affecting strength of a tension member

(c) Steel Splices & Steel Gussets

(d) Types of Tension Member & their neat sketches

(e) Modes of Failure in tension members.

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